

# Development, simulation & implementation of precoding techniques for satellite links

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Ghent, May 2026

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# Acknowledgements

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by

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Faculty of Engineering and Architecture

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**Abstract** — Laboris magna irure quis sit et veniam voluptate duis elit esse do excepteur. Officia laboris adipisicing laboris duis ut cillum aliquip. Do veniam aute non do cillum sit pariatur consectetur adipisicing qui. Lorem Lorem veniam minim exercitation cupidatat tempor do aute dolore. Nostrud adipisicing tempor quis qui fugiat sint quis ut aute eiusmod Lorem enim veniam. Elit labore Lorem do proident amet anim exercitation culpa enim cillum qui. Aliquip reprehenderit commodo labore minim laborum mollit ut veniam in irure. Minim sit in non aliqua ut elit veniam irure pariatur non velit consequat. Cupidatat veniam occaecat culpa eu irure aute consectetur dolore culpa deserunt quis officia do. Aliqua aliqua eu velit excepteur culpa ea dolore occaecat. Ea est cillum nulla ea cupidatat esse adipisicing ut. Cillum amet esse ex ex enim elit excepteur ullamco in qui aliqua Lorem veniam. Sunt consectetur aliquip commodo exercitation pariatur id tempor adipisicing aliqua. Fugiat cillum ea irure laborum consectetur ipsum. Excepteur laboris eu veniam ex ad consequat esse id ullamco ea cupidatat ad nisi. Aute nisi velit excepteur adipisicing dolore pariatur sunt. Dolore mollit cupidatat esse sint ipsum est nulla nisi eu cupidatat excepteur sint cillum. Culpa irure magna culpa eiusmod ut nisi ipsum veniam elit est occaecat veniam. Sunt adipisicing eu minim qui ea laboris ullamco ut. Aliquip occaecat in occaecat ut tempor dolore sunt consequat qui cillum reprehenderit. Voluptate commodo cupidatat ullamco sint aliquip veniam consequat do. Nisi velit excepteur aute laboris aliqua fugiat minim occaecat quis non incididunt.

**Keywords** — Precoding Techniques, High-Troughput Satellite Communications, multiple-input multiple-output (MIMO), imperfect channel state information (CSI)

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TABLE I  
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| Table Head | Table Column Head            |         |         |
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Fig. 1. Example of a figure caption.

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#### ACKNOWLEDGMENT

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#### REFERENCES

- [1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, “Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios,” *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.

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# Symbols and Notations

## Abbreviations

|               |                                         |
|---------------|-----------------------------------------|
| <b>acf</b>    | autocorrelation function                |
| <b>AWGN</b>   | additive white Gaussian noise           |
| <b>BS</b>     | base station                            |
| <b>BD</b>     | block diagonalization                   |
| <b>BER</b>    | bit error rate                          |
| <b>CS</b>     | circularly symmetric                    |
| <b>CSI</b>    | channel state information               |
| <b>DFT</b>    | discrete Fourier transform              |
| <b>DPC</b>    | dirty paper coding                      |
| <b>EVD</b>    | eigenvalue decomposition                |
| <b>FFT</b>    | fast Fourier transform                  |
| <b>FIR</b>    | finite impulse response                 |
| <b>IBR</b>    | information bit rate                    |
| <b>i.i.d.</b> | independent and identically distributed |
| <b>KKT</b>    | Karush-Kuhn-Tucker                      |
| <b>LSV</b>    | left singular vector                    |
| <b>MI</b>     | mutual information                      |
| <b>MIMO</b>   | multiple-input multiple-output          |
| <b>MMSE</b>   | minimum mean squared error              |
| <b>MSE</b>    | mean squared error                      |
| <b>MU</b>     | multi-user                              |
| <b>pdf</b>    | probability density function            |

|              |                                     |
|--------------|-------------------------------------|
| <b>PAM</b>   | pulse amplitude modulation          |
| <b>PSD</b>   | power spectral density              |
| <b>PSK</b>   | phase shift keying                  |
| <b>QAM</b>   | quadrature amplitude modulation     |
| <b>QoS</b>   | Quality-of-Service                  |
| <b>RX</b>    | receiver                            |
| <b>SISO</b>  | single-input single-output          |
| <b>SNR</b>   | signal-to-noise ratio               |
| <b>SR</b>    | sum rate                            |
| <b>SU</b>    | single-user                         |
| <b>SVD</b>   | singular value decomposition        |
| <b>THP</b>   | Tomlinson–Harashima precoding       |
| <b>TX</b>    | transmitter                         |
| <b>UT</b>    | user terminal                       |
| <b>WMSE</b>  | weighted mean squared error         |
| <b>WMMSE</b> | weighted minimum mean squared error |
| <b>WSR</b>   | weighted sum rate                   |
| <b>WSRBF</b> | weighted sum rate beamforming       |
| <b>ZF</b>    | zero-forcing                        |

## Mathematical Notations

TO DO

## System Parameters

|       |                                       |
|-------|---------------------------------------|
| $N_T$ | Number of transmit antennas at the BS |
| $N_R$ | Number of receive antennas at each UT |
| $N_S$ | Number of data streams per UT         |
| $K$   | Number of UTs                         |

## Symbols

TO DO

## Part I

# Background Material

## Chapter 4

# Precoding for MU-MIMO systems

Consider the downlink MU-MIMO system introduced in Section 2.2.2. The key difference from the single-user (SU)-MIMO system of the previous chapter is the presence of inter-user interference: the signal intended for one user leaks into the receive antennas of the other users.

The preceding chapter restricted attention to linear precoding, in which the transmitted signal is obtained by applying a linear transformation, represented by a matrix, to the data symbol vector. Nonlinear precoding techniques, on the other hand, apply nonlinear processing to the data symbols before transmission. While linear precoding can achieve the capacity of the SU-MIMO channel, it is no longer capacity-achieving in the MU-MIMO setting.

Two well-known nonlinear precoding techniques are dirty paper coding (DPC) [16] and Tomlinson–Harashima precoding (THP) [17, 18]. DPC is an information-theoretic technique that achieves the capacity region of the Gaussian broadcast channel by encoding each data stream such that interference from all previously encoded streams is pre-subtracted at the transmitter. THP approximates DPC at reduced complexity by successively canceling inter-user interference at the transmitter using modulo arithmetic. Despite their superior capacity, nonlinear techniques incur a significantly higher computational cost. Linear precoding, by contrast, offers near-optimal performance at a fraction of the complexity, and therefore remains the preferred approach in practice. This chapter accordingly focuses on linear precoding techniques.

The following sections presents four commonly used linear precoding strategies for the MU-MIMO downlink, arranged in order of increasing complexity. The presentation begins with three precoding techniques related to ZF, and concludes with the state-of-the-art WMMSE precoder. At the end of this chapter, their performance is compared through numerical simulations for different system configurations. In line with the previous chapter, the design is focused on maximizing the total sum rate (SR) of the system.

## 4.1 Zero-Forcing Precoding

The objective of zero-forcing (ZF) precoding is to cancel all interference, both inter-user and inter-stream. Moreover, no receiver-side combining is applied. Instead, all processing is done by the precoder and each UT forwards its received samples directly for detection.

Considering these restrictions, we obtain that

$$\mathbf{T} = (\mathbf{G}\mathbf{H}\mathbf{F}) = \sqrt{\mathbf{Q}} \quad \text{and} \quad \mathbf{G} = \mathbf{I}_{\text{rect}}, \quad (4.1)$$

where  $\mathbf{T}$  is the compound MU-MIMO transfer matrix,  $\sqrt{\mathbf{Q}} \in \mathbb{R}_{>0}^{K N_S \times K N_S}$  is a diagonal scaling matrix and  $\mathbf{I}_{\text{rect}}$  is a rectangular diagonal matrix with unit entries on its main diagonal. The compound MU-MIMO input-output relation (2.10) then simplifies to

$$z_{k,i} = q_{k,i} a_{k,i} + n_{k,i} \quad \text{for } k = 1, \dots, K \quad \text{and} \quad i = 1, \dots, N_S. \quad (4.2)$$

This shows that the interference has been completely eliminated and each data stream is received over an independent scalar additive white Gaussian noise (AWGN) channel, which allows performing symbol-by-symbol detection at each UT.

It is shown in [19] that the constraints in (4.1) are satisfied when the precoder is chosen as the right pseudo-inverse of the channel matrix, scaled by  $\sqrt{\mathbf{Q}}$ :

$$\mathbf{F} = \mathbf{H}^H (\mathbf{H}\mathbf{H}^H)^{-1} \sqrt{\mathbf{Q}} \quad (4.3)$$

This expression requires the channel matrix to have full row rank ( $R_H = K N_S$ ), so that the inverse  $(\mathbf{H}\mathbf{H}^H)^{-1}$  exists. For an independent and identically distributed (i.i.d.) Rayleigh fading channel, this condition holds almost surely provided that  $N_T \geq K N_S$ . In what follows, this condition is assumed to be satisfied. If it is not, the inverse  $(\mathbf{H}\mathbf{H}^H)^{-1}$  is replaced by the Moore-Penrose pseudo-inverse, which is computed from the singular value decomposition (SVD) of the channel matrix [20].

It remains to determine the optimal diagonal entries of the scaling matrix  $\mathbf{Q}$  so as to maximize the weighted sum rate (WSR) under the total transmit power constraint. Considering (4.2), the WSR of the system can be written as

$$\text{WSR} = \sum_{k=1}^K \sum_{i=1}^{N_S} \mu_{k,i} \log_2 \left( 1 + \frac{q_{k,i}}{N_0} \right), \quad (4.4)$$

where  $\mu_{k,i}$  is the weight assigned to the  $i$ th data stream intended for UT  $k$ . Substituting the ZF precoder into the transmit power constraint (2.8) yields

$$\sum_{k=1}^K \sum_{i=1}^{N_S} q_{k,i} ((\mathbf{H}\mathbf{H}^H)^{-1})_{n,n} \leq P_t \quad \text{with} \quad n = (k-1)N_S + i. \quad (4.5)$$



Let us introduce the substitution

$$p_n = \gamma_n^{-1} q_{k,i} \quad \text{with} \quad \gamma_n^{-1} = ((\mathbf{H}\mathbf{H}^H)^{-1})_{n,n}, \quad (4.6)$$

such that we can write the WSR maximization problem as follows:

$$\begin{aligned} \max_{\{p_n\}} \quad & \sum_{n=1}^{K N_S} \alpha_n \log_2 \left( 1 + \frac{\gamma_n}{N_0} p_n \right) \\ \text{s.t.} \quad & \sum_{n=1}^{K N_S} p_n \leq P_t \end{aligned} \quad (4.7)$$

This is exactly the optimization problem handled in Appendix A. The optimal values for  $p_i$  are therefore obtained using the waterfilling solution as described in Algorithm 1.

Note that the above power allocation strategy may allocate almost no power to the data streams of a UT with poor channel conditions. The achievable rate for that specific user will therefore be very small, compared to the other users with better channel conditions. An alternative, more fair approach is to divide the total transmit power equally across the different users, and use the waterfilling solution to allocate the power to which every user is entitled across the different data streams intended for that user.

To summarize, the steps required for computing the ZF compound precoder matrix for total SR maximization across all UTs are described in Algorithm 1.

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**Algorithm 1** ZF Precoding

---

**Require:**

- The compound channel matrix  $\mathbf{H}$ ;
- The total transmit power  $P_t$  and the received noise power  $N_0$ .

**Ensure:** The compound precoder matrix  $\mathbf{F}$ .

- 1: Compute  $p_i$  for  $i = 1, \dots, K N_S$  using the waterfilling algorithm (Alg. 1) with inputs  $P_t = P_t$ ,  $\alpha_n = 1$ , and  $\gamma_n = (N_0 ((\mathbf{H}\mathbf{H}^H)^{-1})_{n,n})^{-1}$ .
- 2: Compute the entries of the scaling matrix as  $q_{k,i} = p_n ((\mathbf{H}\mathbf{H}^H)^{-1})_{n,n}$  with  $n = (k-1)N_S + i$ , for  $k = 1, \dots, K$  and  $i = 1, \dots, N_S$ .
- 3: Compute the compound precoder matrix as  $\mathbf{F} = \mathbf{H}^H (\mathbf{H}\mathbf{H}^H)^{-1} \sqrt{\mathbf{Q}}$

**Return**  $\mathbf{F}$

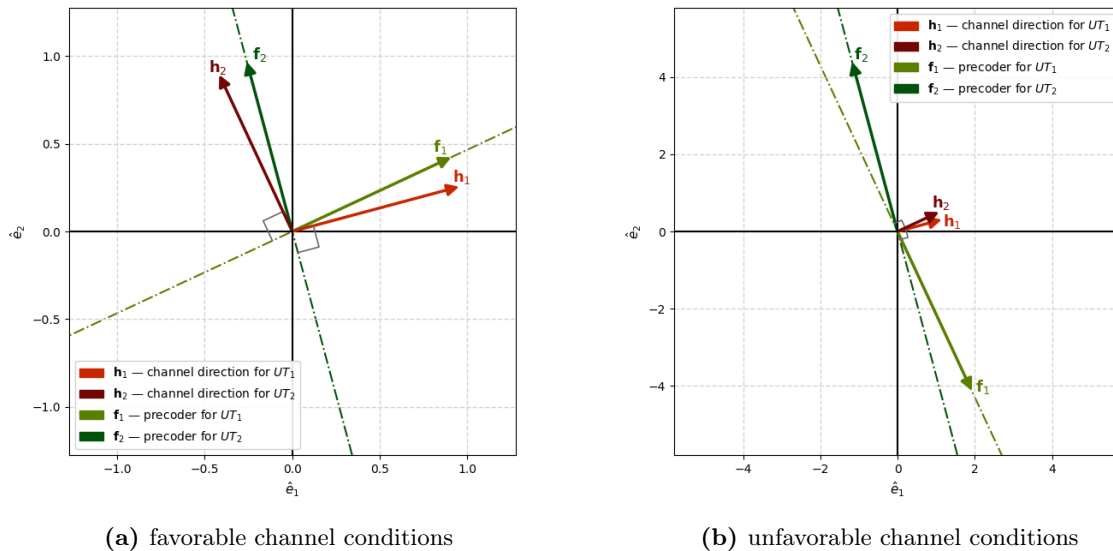
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To build an intuitive understanding of ZF precoding, consider a downlink MU-MIMO system with two single-antenna UTs and a BS with two transmit antennas. In this case, the channel matrix is formed by two row vectors, each describing the channel direction toward one user, while the precoder consists of two column vectors, each defining the transmit direction of the data stream intended for a specific UT. The ZF design chooses each precoding vector to be orthogonal to the channel vector of the non-intended user. This ensures that the signal intended for one UT does not leak into the receive antenna of the other UT, thereby eliminating inter-user interference.

This geometry is illustrated in Figure 4.1, where Figure 4.1a corresponds to favorable channel conditions and Figure 4.1b to unfavorable channel conditions.

In the favorable case, the channel directions for the two UTs are nearly orthogonal. As a result, the ZF precoding vectors can remain close to the intended channel directions, which preserves a high effective channel gain. In the unfavorable case, the channel directions are strongly correlated (nearly parallel), so the ZF constraints force the precoding vectors away from the desired channel directions. This substantially reduces the effective gain.

For a fair geometric comparison, the channel vectors in both figures are normalized, and the precoding vectors are scaled such that each stream attains unit effective channel gain. Under this normalization, the much larger precoder norms in the unfavorable case indicate a significantly higher transmit-power requirement to achieve the same effective gain. With a fixed total transmit power in practice, this translates directly into lower received signal-to-noise ratio (SNR) and therefore a lower achievable SR.

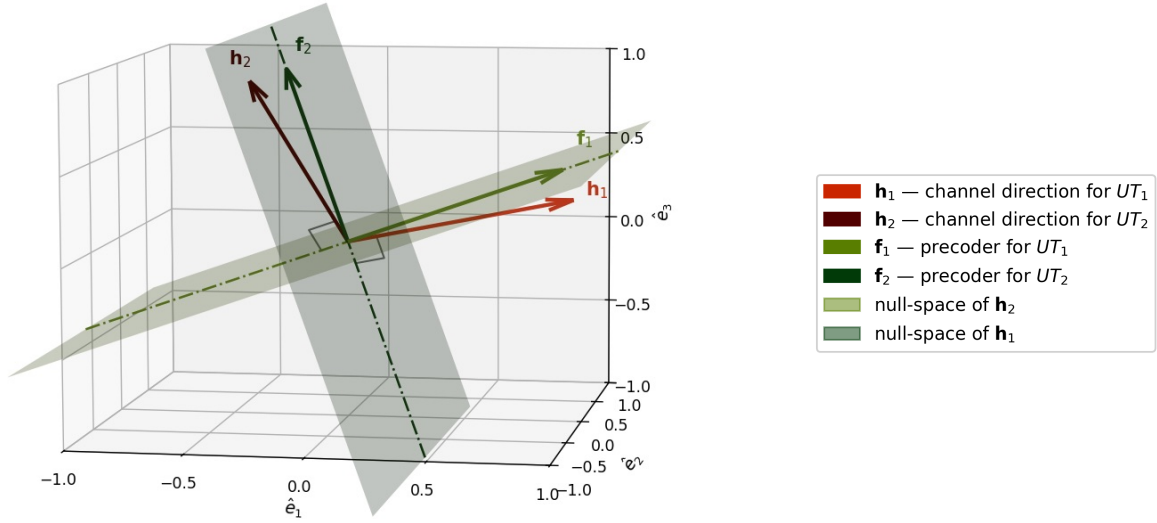


**Figure 4.1:** Visualization of ZF precoder for a MU-MIMO system with 2 single-antenna UTs and a 2-antenna BS under (a) favorable and (b) unfavorable channel conditions. For both scenarios, the channel vectors are normalized and the precoding vectors are scaled such that the effective channel gain for each data stream is equal to 1. Note the much larger norm of the precoding vectors in the unfavorable scenario compared to the favorable scenario, which illustrates that a lot more power is required to achieve the same effective channel gain when the channel vectors of the UTs are strongly correlated.

Figure 4.2 illustrates what happens when the BS is equipped with an additional transmit antenna while the number of single-antenna UTs remains equal to two. The channel matrix is still formed by two row vectors, one for each user, and the precoder by two column vectors. However, these vectors now lie in a three-dimensional space. As a result, the null-space orthogonal to each user's channel direction is no longer a line but a two-dimensional plane.

In order to eliminate all interference, the precoding vector for each user can be chosen anywhere in the null-space of the other users' channel vectors, that is, in the plane orthogonal to the channel directions of the other users. The best choice is to select the precoding vectors as close as possible to the intended channel directions, which corresponds to projecting the channel vectors onto the

respective orthogonal planes. This maximizes the effective channel gain for each data stream, since it allows as much power as possible to be transmitted in the direction of the intended UT.



**Figure 4.2:** Visualization of ZF precoder for a MU-MIMO system with 2 single-antenna UTs and a 3-antenna BS. The channel vectors are normalized and the precoding vectors are scaled such that the effective channel gain for each data stream is equal to 1. With three transmit antennas, the null-space orthogonal to each user's channel direction is two-dimensional. The ZF precoding vector is chosen as the projection of the intended user's channel direction onto this plane, which minimizes the deviation from the desired channel direction while preserving the zero-forcing constraint.

## 4.2 Zero-Forcing Precoding with an Heuristic LSV Combiner

Pure ZF precoding is clearly suboptimal because it does not exploit receiver-side combining at the UTs. A practical extension is therefore to equip each UT with a heuristic left singular vector (LSV) combiner, while keeping the BS precoder design based on ZF. The key idea is to improve the effective channel gains seen without increasing the signaling overhead. Unlike coordinated designs, where the BS jointly computes the compound precoder and the combiners and then broadcasts the combiners to the UTs, this approach is non-coordinated: each UT computes its own combiner locally.

The procedure is as follows. First, the BS transmits pilot symbols, from which each UT estimates its downlink channel matrix  $\mathbf{H}_k$ . Next, each UT constructs its heuristic combiner matrix  $\mathbf{G}_k$  from the  $N_S$  dominant left singular vectors of  $\mathbf{H}_k$ , analogous to combiner for capacity-achieving precoding in the SU-MIMO scenario. Each UT then forms its virtual channel ( $\mathbf{G}_k \mathbf{H}_k$ ) and feeds this matrix back to the BS. Finally, the BS stacks these virtual channels and applies the standard ZF precoding procedure from the previous section on the virtual channel.

This method was proposed by Sander Cornelis in [8] and is particularly attractive in feedforward-limited scenarios. The complete procedure is summarized in Algorithm 2.

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**Algorithm 2** ZF Precoding with Heuristic LSV combiner
 

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**Require:**

- The compound channel matrix  $\mathbf{H}$ ;
- The total transmit power  $P_t$  and the received noise power  $N_0$ .

**Ensure:** The compound precoder matrix  $\mathbf{F}$  and the combiner matrices  $\mathbf{G}_k$  for each UT.

 1: Compute the combiner matrices  $\mathbf{G}_k$  for each UT as

$$\mathbf{G}_k = [\mathbf{u}_{k,1} \ \dots \ \mathbf{u}_{k,N_S}]^H, \quad \text{with} \quad \mathbf{H}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^H$$

 2: Compute the compound precoder matrix  $\mathbf{F}$  as using the ZF algorithm (Alg. 1) with inputs

$$\mathbf{H} = (\mathbf{G}\mathbf{H}), \quad P_t = P_t, \quad N_0 = N_0.$$

**Return**  $\mathbf{F}$ ,  $\mathbf{G}_k$  for  $k = 1, \dots, K$ 


---

### 4.3 Block Diagonalization Precoding

The last precoding technique in the class of ZF precoders is a coordinated technique called block diagonalization (BD) precoding [21]. In contrast to pure ZF, which cancels both inter-user and inter-stream interference, BD only suppresses the inter-user interference. As a result, the compound transfer matrix is no longer required to be diagonal, but only block-diagonal, so that each UT receives a signal vector free of contributions of data streams intended for other UTs. The downlink MU-MIMO channel is thereby decomposed into  $K$  parallel SU-MIMO channels. Once this first step is completed, the SU-MIMO precoding and combining techniques developed in Chapter 3 can be applied independently to each subsystem.

The zero inter-user interference condition requires that the precoder  $\mathbf{F}_k$  intended for UT  $k$  lie in the null space of the channel matrices of all other users. This leads to the constraints

$$\forall k' \neq k : \quad \mathbf{H}_{k'} \mathbf{F}_k = \mathbf{0}, \quad \text{for } k = 1, \dots, K. \quad (4.8)$$

To characterize these constraints more explicitly, define the interfering channel matrix  $\mathring{\mathbf{H}}_k$  of UT  $k$  as the vertically stacked channel matrices of all other users:

$$\mathring{\mathbf{H}}_k = [\mathbf{H}_1^H \ \dots \ \mathbf{H}_{k-1}^H \ \mathbf{H}_{k+1}^H \ \dots \ \mathbf{H}_K^H]^H \in \mathbb{C}^{(K-1)N_R \times N_T}. \quad (4.9)$$

The constraint in (4.8) is then equivalent to forcing the transmit directions for UT  $k$  to lie within the null-space of its interfering channel matrix.

To obtain a basis for this null space, consider the SVD of the interfering channel matrix:

$$\mathring{\mathbf{H}}_k = \mathring{\mathbf{U}}_k \mathring{\mathbf{\Sigma}}_k \begin{bmatrix} \mathring{\mathbf{V}}_k^{(1)} & \mathring{\mathbf{V}}_k^{(0)} \end{bmatrix}^H. \quad (4.10)$$

The first  $R_{\mathring{H}_k}$  right singular vectors of  $\mathring{\mathbf{H}}_k$ , collected in  $\mathring{\mathbf{V}}_k^{(1)}$ , span the row space of  $\mathring{\mathbf{H}}_k$ , whereas the remaining  $(N_T - R_{\mathring{H}_k})$  right singular vectors, collected in  $\mathring{\mathbf{V}}_k^{(0)}$ , span its null space. By  $R_{\mathring{H}_k}$  we denote the rank of the interfering channel matrix  $\mathring{\mathbf{H}}_k$ . Therefore, any precoder of the form

$$\mathbf{F}_k^{\text{MU}} = \mathring{\mathbf{V}}_k^{(0)} \quad (4.11)$$

automatically satisfies the zero inter-user interference condition.

Data transmission to UT  $k$  is only possible if the null space of  $\mathring{\mathbf{H}}_k$  is non-empty, i.e. it has a dimension greater than zero. This condition is satisfied if  $R_{\mathring{H}_k} < N_T$ , which is guaranteed under the standing assumption that the number of transmit antennas is greater than or equal to the total number of receive antennas across all users, i.e.  $N_T \geq KN_R$ . In addition, the channel as seen after the projection of  $\mathbf{F}_k^{\text{MU}}$ , i.e.  $(\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)})$ , must still retain useful spatial dimensions. In other words, the channel directions of UT  $k$  should not be entirely contained in the row space of the interfering UTs. A sufficient condition is that at least one row of  $\mathbf{H}_k$  is linearly independent of the rows of  $\mathring{\mathbf{H}}_k$  [22]. This shows that BD, like ZF, benefits from users with sufficiently distinct spatial signatures, i.e. UTs without highly correlated channel directions. Note that pure ZF imposes the stronger requirement that all rows of  $\mathbf{H}_k$  are linearly independent of  $\mathring{\mathbf{H}}_k$ . While this is not necessary for BD, it would result in a higher rank of  $(\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)})$ , and therefore greater degrees of freedom in the subsequent SU-MIMO design step.

If, in a first step, the receiver combiners are ignored and the multi-user precoder is chosen as  $\mathbf{F}_k^{\text{MU}} = \mathring{\mathbf{V}}_k^{(0)}$ , then the compound MU-MIMO input-output relation simplifies to

$$\mathbf{y}_k = (\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)}) \mathbf{a}_k + \mathbf{n}_k, \quad \text{for } k = 1, \dots, K. \quad (4.12)$$

Hence, the original downlink MU-MIMO system is transformed into  $K$  parallel SU-MIMO subsystems, each with  $(N_T - R_{\mathring{H}_k})$  effective transmit antennas,  $N_R$  receive antennas, and an effective channel matrix  $\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)}$ . The second step is then to apply a SU-MIMO precoder and combiner design to each of these effective channels. In this dissertation, the focus is on maximum-

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**Algorithm 3** BD Precoding

---

**Require:**

- The compound channel matrix  $\mathbf{H}$ ;
- The total transmit power  $P_t$  and the received noise power  $N_0$ .

**Ensure:** The compound precoder matrix  $\mathbf{F}$  and the combiner matrices  $\mathbf{G}_k$  for each UT.

- 1: Construct the interfering channel matrix for each UT  $k$  as  
 $\mathring{\mathbf{H}}_k = [\mathbf{H}_1^H \ \dots \ \mathbf{H}_{k-1}^H \ \mathbf{H}_{k+1}^H \ \dots \ \mathbf{H}_K^H]^H \in \mathbb{C}^{(K-1)N_R \times N_T}$ .
- 2: Compute the MU precoder matrix for each UT  $k$  as  
 $\mathbf{F}_k^{\text{MU}} = \mathring{\mathbf{V}}_k^{(0)}$ , with  $\mathring{\mathbf{H}}_k = \mathring{\mathbf{U}}_k \mathring{\mathbf{\Sigma}}_k \begin{bmatrix} \mathring{\mathbf{V}}_k^{(1)} & \mathring{\mathbf{V}}_k^{(0)} \end{bmatrix}^H$ .
- 3: Compute the SU precoder and combiner matrix for each UT  $k$  as  
 $\mathbf{F}_k^{\text{SU}} = [\bar{\mathbf{v}}_{k,1} \ \dots \ \bar{\mathbf{v}}_{k,N_S}]$  and  $\mathbf{G}_k = [\bar{\mathbf{u}}_{k,1} \ \dots \ \bar{\mathbf{u}}_{k,N_S}]^H$ , with  $(\mathbf{H}_k \mathring{\mathbf{V}}_k^{(0)}) = \bar{\mathbf{U}}_k \bar{\mathbf{\Sigma}}_k \bar{\mathbf{V}}_k^H$ .
- 4: Extract the eigen subchannel gains as  
 $\sigma_n = (\bar{\mathbf{\Sigma}}_k)_{i,i}$  with  $n = (k-1)N_S + i$ , for  $k = 1, \dots, K$  and  $i = 1, \dots, N_S$ .
- 5: Compute the optimal power allocation  $p_i$  using the waterfilling algorithm (Alg. 1) with inputs  
 $P_t = P_t$ ,  $\alpha_n = 1$ , and  $\gamma_n = \sigma_n^2/N_0$ .
- 6: Form the compound precoder matrix as  
 $\mathbf{F} = [\mathbf{F}_1^{\text{MU}} \mathbf{F}_1^{\text{SU}} \ \dots \ \mathbf{F}_K^{\text{MU}} \mathbf{F}_K^{\text{SU}}] \sqrt{\mathbf{P}}$  with  $\mathbf{P} = \text{diag}(p_1, \dots, p_{KN_S})$ .

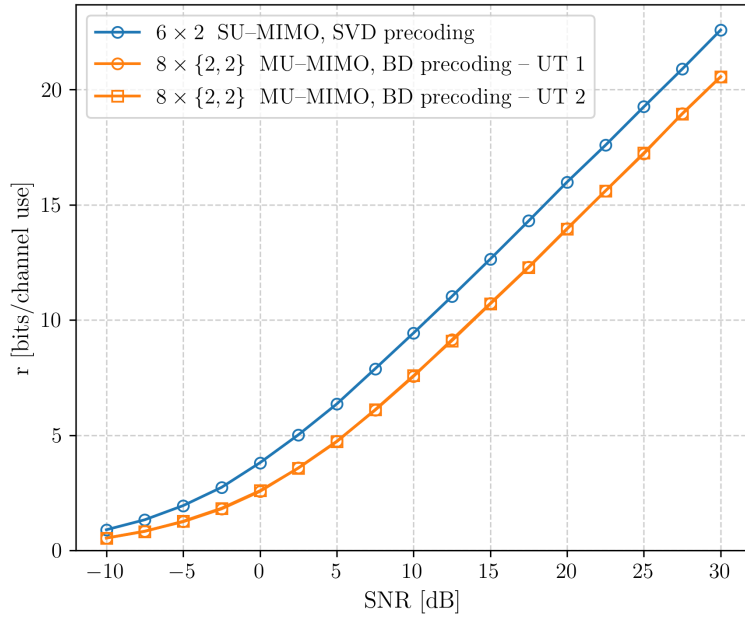
**Return**  $\mathbf{F}$ ,  $\mathbf{G}_k$  for  $k = 1, \dots, K$

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information-rate design. Therefore, we adapt the capacity-achieving precoding technique from Section 3.1.2, so that each effective channel is diagonalized by means of its SVD, followed by a global waterfilling power allocation across all resulting eigen subchannels. The full BD procedure is summarized in Algorithm 3.

As with pure ZF, the chosen power allocation may devote very little power to users whose effective channels are weak. A more balanced alternative is to divide the total transmit power equally across the users and then apply waterfilling separately over the eigen subchannels of each SU-MIMO subsystem. More generally, the second stage of the BD design need not be based on maximum sum-rate. Other SU-MIMO design criteria, such as Quality-of-Service (QoS)-based or minimum mean squared error (MMSE)-based precoding, can equally be applied to the parallel channels obtained after the block-diagonalization step.

To conclude this section, Figure 4.3 compares the achievable-rate performance of BD precoding in a downlink MU-MIMO setting with the capacity-achieving SVD precoding benchmark in a corresponding SU-MIMO setting. The reference system is a  $6 \times 2$  SU-MIMO link, whereas the multi-user system is an  $8 \times \{2, 2\}$  MU-MIMO link with two UTs, each equipped with two receive antennas. The BD algorithm decouples the MU-MIMO system into 2 parallel SU-MIMO subsystems with, on average, each 6 effective transmit antennas and 2 receive antennas. The rates are averaged over 1000 i.i.d. Rayleigh channel realizations.



**Figure 4.3:** Comparison between the achievable rate of a  $6 \times 2$  SU-MIMO system using SVD precoding and the achievable rates of both UTs in an  $8 \times \{2, 2\}$  MU-MIMO system using BD precoding. Results are averaged over 1000 i.i.d. Rayleigh channel realizations. The SNR is defined as the ratio between the total transmit power  $P_t$  at the BS and the noise power per receive antenna  $N_0$ .

The curves show that the achievable-rate slope as a function of SNR is essentially the same for all users. In addition, the SU benchmark exhibits only a small SNR advantage over each user in the multi-user (MU) scenario. The two users in the MU scenario achieve nearly identical rates,

which is expected because the MU system is statistically symmetric with respect to the users and the results are averaged over multiple channel realizations.

This comparison indicates that, on average, adding two transmit antennas at the BS and switching from SVD precoding in SU-MIMO to BD precoding in MU-MIMO enables serving an additional two-stream user with the same reliability only a minor per-user throughput penalty. From a system-design perspective, this highlights the practical spatial multiplexing capability of BD precoding. More generally, as more UTs are served simultaneously, the slope of the total SR versus SNR increases, at the cost of a modest reduction in the achievable rate of each individual UT.

## 4.4 Weighted Minimum Mean Square Error Precoding

The three precoding techniques discussed so far all enforce hard structural constraints on the precoders to handle the interference. Pure ZF and ZF with an LSV combiner cancel it completely through a channel inversion, while BD confines each user's precoder to the null space of the other users' channels. In both cases, satisfying these structural constraints prevents the precoder from operating in directions that could simultaneously offer high signal gain and tolerable interference, and thus limits the achievable performance. None of these designs directly maximizes the weighted sum rate (WSR) of the system. In this section, we present a precoding technique with the explicit goal of maximizing the WSR under a total transmit power constraint.

Consider the WSR maximization problem:

$$\max_{\{\mathbf{F}_k\}} \sum_{k=1}^K w_k R_k \quad \text{s.t.} \quad \sum_{k=1}^K \text{tr}(\mathbf{F}_k \mathbf{F}_k^H) \leq P_t. \quad (4.13)$$

Unlike its single-user counterpart in Chapter 3, problem (4.13) is non-convex in  $\{\mathbf{F}_k\}$  and admits no closed-form solution. The state-of-the-art approach, proposed by the authors of [23], is to recast it as an equivalent weighted minimum mean squared error (WMMSE) problem and to solve the latter by alternating optimization.

Consider the system input–output relation for UT  $k$  from (2.9). The signals intended for the other users act, under Gaussian signaling, as additional Gaussian noise with covariance matrix

$$\mathbf{C}_{\tilde{n}_k} = N_0 \mathbf{I}_{N_R} + \sum_{k' \neq k} \mathbf{H}_k \mathbf{F}_{k'} \mathbf{F}_{k'}^H \mathbf{H}_k^H. \quad (4.14)$$

Consequently, the achievable rate of UT  $k$  can be written similarly to the SU case as

$$R_k = \log_2 \det \left( \mathbf{I}_{N_S} + \mathbf{F}_k^H \mathbf{H}_k^H \mathbf{C}_{\tilde{n}_k}^{-1} \mathbf{H}_k \mathbf{F}_k \right). \quad (4.15)$$

For a given set of precoders, the minimum mean squared error (MMSE) combiner of UT  $k$  is the Wiener filter that minimizes the mean–square estimation error in the presence of inter–user

interference and noise, and is given by [24]:

$$\mathbf{G}_k^{\text{MMSE}} = \mathbf{F}_k^H \mathbf{H}_k^H (\mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^H \mathbf{H}_k^H + \mathbf{C}_{\tilde{n}_k})^{-1}. \quad (4.16)$$

The corresponding mean squared error (MSE) matrix, defined as in Section 3.2.1 but specialized to UT  $k$ , then reduces to [25]

$$\mathbf{E}_k = \mathbb{E}[(\mathbf{a}_k - \mathbf{z}_k)(\mathbf{a}_k - \mathbf{z}_k)^H] = (\mathbf{I}_{N_S} + \mathbf{F}_k^H \mathbf{H}_k^H \mathbf{C}_{\tilde{n}_k}^{-1} \mathbf{H}_k \mathbf{F}_k)^{-1}. \quad (4.17)$$

Comparing (4.15) and (4.17), the achievable rate of UT  $k$  can be expressed directly in terms of its MMSE matrix as [23]

$$R_k = \log_2 \det(\mathbf{E}_k^{-1}). \quad (4.18)$$

This identity is the key bridge between the WSR maximization problem and the WMMSE minimization problem. It is the multi-user analogue of the rate–MSE link already established in the generalized WMMSE design of Section 3.2.1.

Define the WMMSE optimization problem, in analogy to its SU counterpart in (3.15), as

$$\min_{\{\mathbf{F}_k\}} \sum_{k=1}^K \text{tr}(\mathbf{W}_{e,k} \mathbf{E}_k) \quad \text{s. t.} \quad \sum_{k=1}^K \text{tr}(\mathbf{F}_k \mathbf{F}_k^H) \leq P_t, \quad (4.19)$$

where each  $\mathbf{W}_{e,k} \in \mathbb{C}^{N_S \times N_S}$  is a positive-definite weight matrix associated with UT  $k$  and  $\mathbf{E}_k$  is the MMSE matrix in (4.17). Whereas in the SU case the weight matrix could be chosen a priori so as to recover the maximum-information-rate design (cf. Section 3.2.2), the MU setting is more involved. Comparing the gradients of the Lagrangians of optimization problems (4.13) and (4.19) with respect to  $\mathbf{F}_k$  shows that they coincide whenever [23]

$$\mathbf{W}_{e,k} = w_k \mathbf{E}_k^{-1} \quad \text{for } k = 1, \dots, K. \quad (4.20)$$

At any stationary point of the WSR problem, computing  $\{\mathbf{E}_k\}$  from (4.17) and selecting the weights as in (4.20) therefore yields a stationary point of the WMMSE problem with the same precoders, and vice versa. This equivalence is the MU counterpart of the special case  $\mathbf{W}_e = \mathbf{A}$  identified in Section 3.2.2 for the SU channel. However, in the MU case, the weights depend on the precoders themselves through  $\mathbf{E}_k$ , because the precoder of every user influences the interference experienced by all other users. The WSR problem can therefore not be solved by a single WMMSE design with fixed weights, but must instead be tackled iteratively by alternating between updating the weights and solving the WMMSE problem.

The WMMSE algorithm exploits the equivalence above by cycling, at each iteration, through three closed-form updates. First, the MMSE combiners are updated according to (4.16). Second, the weights are updated according to (4.20). Third, the compound precoder update solves the WMMSE problem (4.19) for fixed combiners and weights, which admits the closed-form solution [26]

$$\bar{\mathbf{F}} = \left( \mathbf{H}^H \mathbf{G}^H \mathbf{W} \mathbf{G} \mathbf{H} + \text{tr}(\mathbf{W} \mathbf{G} \mathbf{G}^H) \frac{N_0}{P_t} \mathbf{I}_{N_T} \right)^{-1} \mathbf{H}^H \mathbf{G}^H \mathbf{W}, \quad (4.21)$$



followed by the rescaling  $\mathbf{F} = \beta \bar{\mathbf{F}}$  with  $\beta^2 = P_t / \text{tr}(\bar{\mathbf{F}}\bar{\mathbf{F}}^H)$  to enforce the total power constraint with equality. The full procedure is summarized in Algorithm 4.

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**Algorithm 4** WMMSE Precoding

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**Require:**

- The compound channel matrix  $\mathbf{H}$ ;
- The total transmit power  $P_t$  and the received noise power  $N_0$ .

**Ensure:** The compound precoder matrix  $\mathbf{F}$  and the combiner matrices  $\mathbf{G}_k$  for each UT.

1: Initialize the precoders as

$$\mathbf{F}_k = \mathbf{F}_k^{(0)} \quad \text{with} \quad \mathbf{F}_k^{(0)} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}) \quad \text{for} \quad k = 1, \dots, K.$$

2: **repeat**

3: Update the combiners as

$$\mathbf{G}_k = \mathbf{F}_k^H \mathbf{H}_k^H (\mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^H \mathbf{H}_k^H + \mathbf{C}_{\tilde{n}_k})^{-1} \quad \text{for} \quad k = 1, \dots, K.$$

4: Update the weights as

$$\mathbf{W}_k = \mathbf{E}_k^{-1} \quad \text{for} \quad k = 1, \dots, K.$$

5: Update the precoder as

$$\begin{aligned} \bar{\mathbf{F}} &= \left( \mathbf{H}^H \mathbf{G}^H \mathbf{W} \mathbf{G} \mathbf{H} + \text{tr}(\mathbf{W} \mathbf{G} \mathbf{G}^H) \frac{N_0}{P_t} \mathbf{I}_{N_T} \right)^{-1} \mathbf{H}^H \mathbf{G}^H \mathbf{W} \\ \mathbf{F} &= \beta \bar{\mathbf{F}} \quad \text{with} \quad \beta^2 = P_t / \text{tr}(\bar{\mathbf{F}}\bar{\mathbf{F}}^H) \end{aligned}$$

6: **until** convergence

**Return**  $\mathbf{F}$ ,  $\mathbf{G}_k$  for  $k = 1, \dots, K$ .

---

It is shown in [23] that the WMMSE precoder converges to a local optimum of the WSR problem. Because the problem is non-convex, the quality of the obtained solution depends on the initialization  $\{\mathbf{F}_k^{(0)}\}$ . Furthermore, the authors pointed out that for any set of precoders, a rate-equivalent set can be constructed such that the MSE matrices are diagonal for all users. This justifies the restriction of the weight update to the diagonal elements, and implies that each UT can detect its  $N_S$  data streams independently. The implementation used throughout this work is this diagonal-restricted version. It runs 10 iterations for 2 different random initializations and retains the precoder that yields the highest WSR.

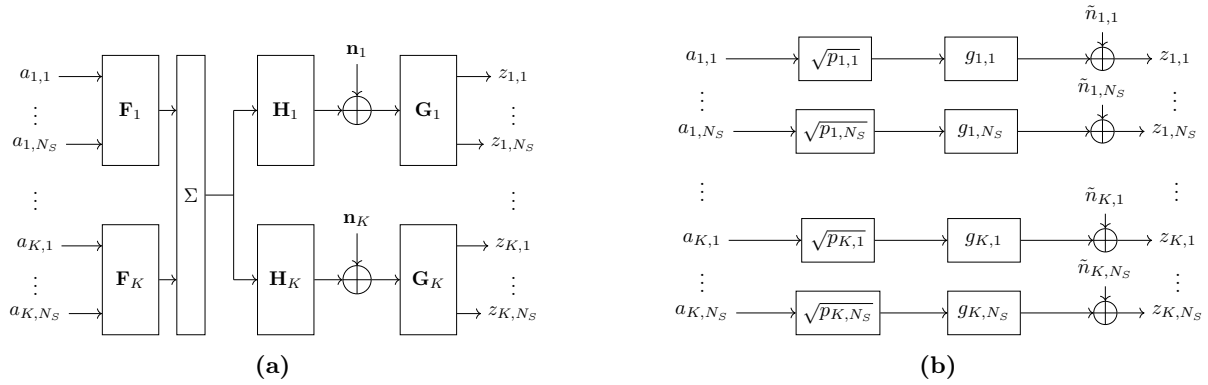
## 4.5 Performance Evaluation

The previous sections derived linear transceiver designs for downlink MU-MIMO sum-rate maximization under different structural constraints. This section evaluates their behavior through Monte Carlo simulations. First, the three ZF-type methods are compared through the statistics of their effective subchannel gains. Second, the achievable SR of all four linear precoding techniques is compared for two different system configurations.

The simulations are conducted under the same assumptions as in Chapter 3: the channel is modeled as an i.i.d. Rayleigh block fading channel (see Appendix C), the reported performance metrics are averaged over at least 100 independent channel realizations, and perfect CSI is assumed at both the BS and the UTs.

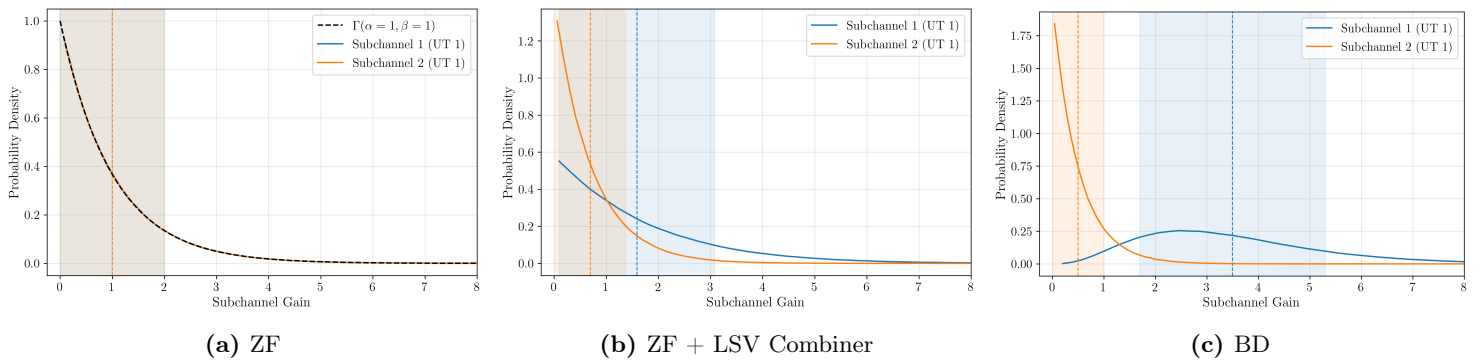
### 4.5.1 Comparison of the ZF Precoders

This subsection compares the three ZF-type designs under the maximum-information-rate criterion: pure ZF, ZF with heuristic LSV combining, and BD. In all three cases, the end-to-end processing transforms the original MU-MIMO channel into parallel scalar Gaussian subchannels, as illustrated in Figure 4.4. Hence, for a fixed power budget and waterfilling policy, performance is completely determined by the statistics of the effective subchannel gains  $g_n$ . The stronger these effective channel gains, the better the overall system performance.



**Figure 4.4:** A block diagram of the original MU-MIMO channel (a) and its decomposition into parallel Gaussian single-input single-output (SISO) subchannels (b), as achieved by the ZF-type precoding techniques.

Figure 4.5 reports the marginal probability density functions (pdfs) for a single user in an system with 8 transmit antennas at the BS and 4 two-antenna UTs, obtained through Monte-Carlo simulations. Because users with the same number of receive antennas are statistically symmetric, this user is representative for the full set of users. For pure ZF, the effective subchannel gains are given by  $g_n = ((\mathbf{H}\mathbf{H}^H)^{-1})_{n,n}^{-1}$  and, under an i.i.d. Rayleigh fading model, their distribution follows from the inverse-Wishart model [27, 28]. In particular, each  $g_n$  is Gamma-distributed with shape parameter  $\alpha = N_T - KN_R + 1$  and rate parameter  $\beta = 1$ . For ZF with an heuristic LSV combiner and for BD precoding, no analytical distribution is derived..



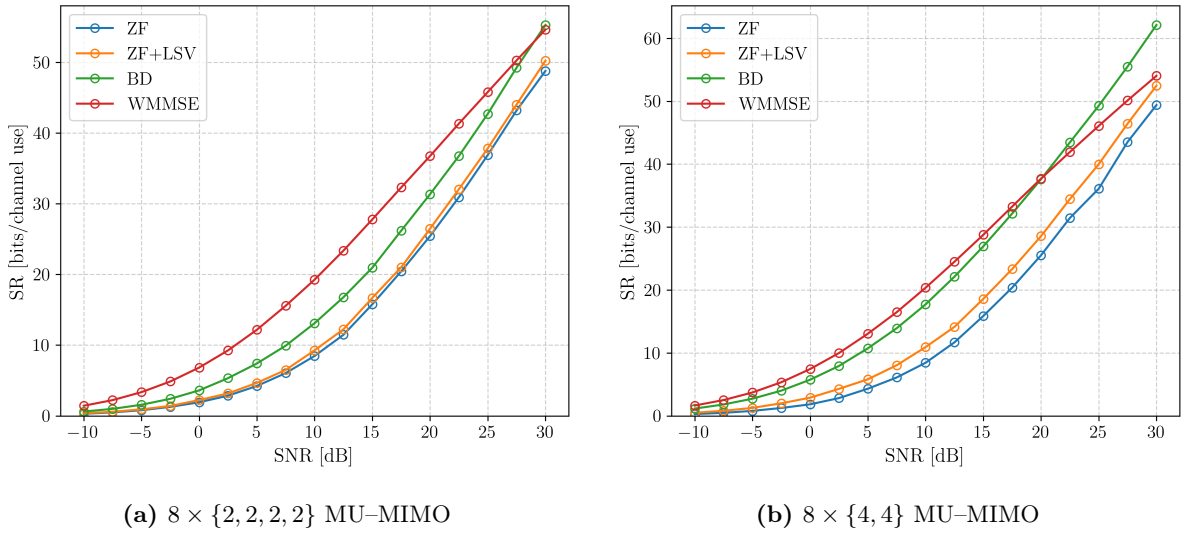
**Figure 4.5:** The marginal pdfs of the effective subchannel gains for a single UT in a  $8 \times \{2, 2, 2, 2\}$  MU-MIMO system, compared between the different ZF-type precoding techniques. The expected value is indicated by a vertical dashed line, and the shaded area around the expected value corresponds to the standard deviation.

For pure ZF (Fig. 4.5a), the two subchannels of a single UT have identical marginal distributions, consistent with the analytical result. Adding the heuristic LSV combiner (Fig. 4.5b) breaks this symmetry between the two subchannels. The first combiner vector is aligned with the dominant left singular vector of  $\mathbf{H}_k$ , which concentrates the available receive-side spatial gain onto the dominant subchannel at the expense of the weaker one. This creates a slightly more favorable result compared to pure ZF. BD precoding (Fig. 4.5c) achieves substantially larger gains for the dominant subchannel than those observed under the ZF with LSV combiner technique. The sum of the expected subchannel gains is clearly the largest, due to the better utilization of the spatial degrees of freedom and the coordinated design. Similar observations hold for other antenna configurations.

These differences in subchannel gain statistics translate directly into differences in achievable sum rate, as will be shown in the next section.

#### 4.5.2 Achievable Sum Rate Comparison

Figure 4.6 compares the achievable SR of all four precoding techniques for two fully loaded MU-MIMO configurations, where “fully loaded” means that the total number of receive antennas equals the number of transmit antennas ( $KN_R = N_T$ ). The two configurations differ only in how the receive antennas are distributed across UTs: four UTs with two receive antennas each in Fig. 4.6a and two UTs with four receive antennas each in Fig. 4.6b.



**Figure 4.6:** Comparison of the achievable sum rate for the considered MU linear precoding techniques in two fully loaded MU-MIMO systems. The results are averaged over at least 1000 i.i.d. Rayleigh channel realizations.

In both configurations, the ranking

$$SR_{\text{WMMSE}} > SR_{\text{BD}} > SR_{\text{ZF+LSV}} > SR_{\text{ZF}}$$

holds at moderate SNR, consistent with the increasing level of design flexibility across the four methods. Pure ZF imposes the strictest constraints, requiring the precoder to invert the full compound channel matrix with no receiver-side processing. ZF with LSV combining retains the

same precoder structure but introduces receive-side spatial processing. BD further relaxes the structural constraints from per-stream to per-user orthogonality and exploits receiver combining through a coordinated design, yielding a substantial gain in achievable SR. WMMSE places no structural constraint on the precoder and directly optimizes the WSR, achieving the highest SR at moderate SNR. The performance differences manifest as a horizontal SNR offset rather than a change in growth rate, because all four techniques achieve the same multiplexing gain.

A notable exception to the above ranking occurs at very high SNR in the  $8 \times \{4, 4\}$  configuration (Figure 4.6b), where BD eventually overtakes WMMSE. This crossover arises because BD approaches the globally optimal solution for the maximum-information-rate criterion as SNR grows [21], while WMMSE may converge to a suboptimal local optimum in this regime.

Comparing the two configurations also reveals how the distribution of receive antennas affects performance. For pure ZF, the two curves are virtually identical: since no receiver combining is applied and the total number of receive antennas is the same, the achievable SR depends only on the compound channel and is unaffected by how the antennas are grouped across UTs. For ZF with LSV combining, BD and WMMSE, concentrating the receive antennas on fewer UTs yields a consistently higher SR. The benefit of concentrating antennas is more pronounced for BD than for ZF+LSV, which reflects the fact that the coordinated BD design makes fuller use of the per-user spatial degrees of freedom than the non-coordinated LSV heuristic.

The simulation results confirm the position of WMMSE as the state-of-the-art linear MU-MIMO precoding technique under the maximum-SR criterion. BD provides a close-to-optimal coordinated alternative that does not require iterative optimization, while ZF+LSV offers a low-overhead non-coordinated baseline. The choice between these techniques in practice therefore involves a tradeoff between achievable SR, computational complexity, and signaling overhead. In the next chapters, performance evaluations are carried out using the WMMSE precoder.

## Part II

# Precoding for Satellite Communication Systems

## Part III

# Appendices

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